Experiment:4

Generation and Analysis of Frequency Modulated (FM) and of Phase Modulated (PM) Signals using MATLAB

Aim: To generate Frequency Modulated (FM) and Phase Modulated (PM) signals using MATLAB. To analyze their time-domain waveforms, frequency spectra, and modulation indices.

Software Required: MATLAB

Program:

```
clc; clear; close all;

Fs=100000; t=0:1/Fs:0.02-1/Fs;
Ac=5; Am=1; fc=10000; fm=500;
df=2000; Kp=2;

msg=Am*cos(2*pi*fm*t);
c=Ac*cos(2*pi*fc*t);
FM=Ac*cos(2*pi*fc*t+(df/fm)*sin(2*pi*fm*t));
PM=Ac*cos(2*pi*fc*t+Kp*msg);

ffm=fc+df*cos(2*pi*fm*t);
fpm=fc-Kp*Am*fm*sin(2*pi*fm*t);

n=round(0.005*Fs);
figure;
subplot(4,1,1); plot(t(1:n),msg(1:n),'b'); grid on;
title('Message Signal'); ylabel('Amplitude (V)');

subplot(4,1,2); plot(t(1:n),c(1:n),'k'); grid on;
title('Carrier Signal'); ylabel('Amplitude (V)');

subplot(4,1,3); plot(t(1:n),FM(1:n),'r'); grid on;
title('FM Signal'); ylabel('Amplitude (V)');

subplot(4,1,4); plot(t(1:n),PM(1:n),'m'); grid on;
title('PM Signal'); xlabel('Time (s)'); ylabel('Amplitude (V)');

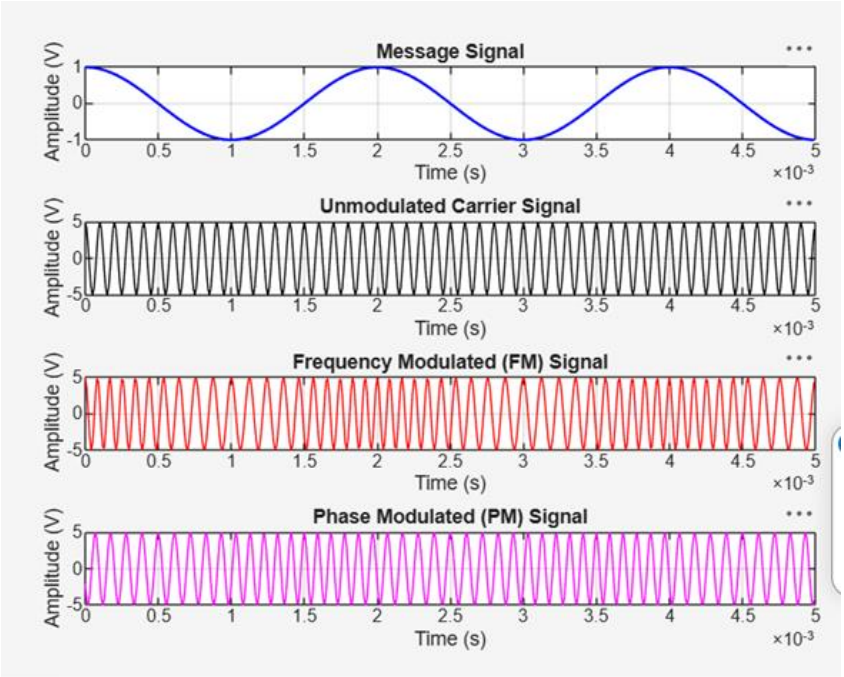
figure;
subplot(2,1,1); plot(t(1:n),ffm(1:n),'r'); grid on;
title('Instantaneous Frequency of FM'); ylabel('Frequency (Hz)');

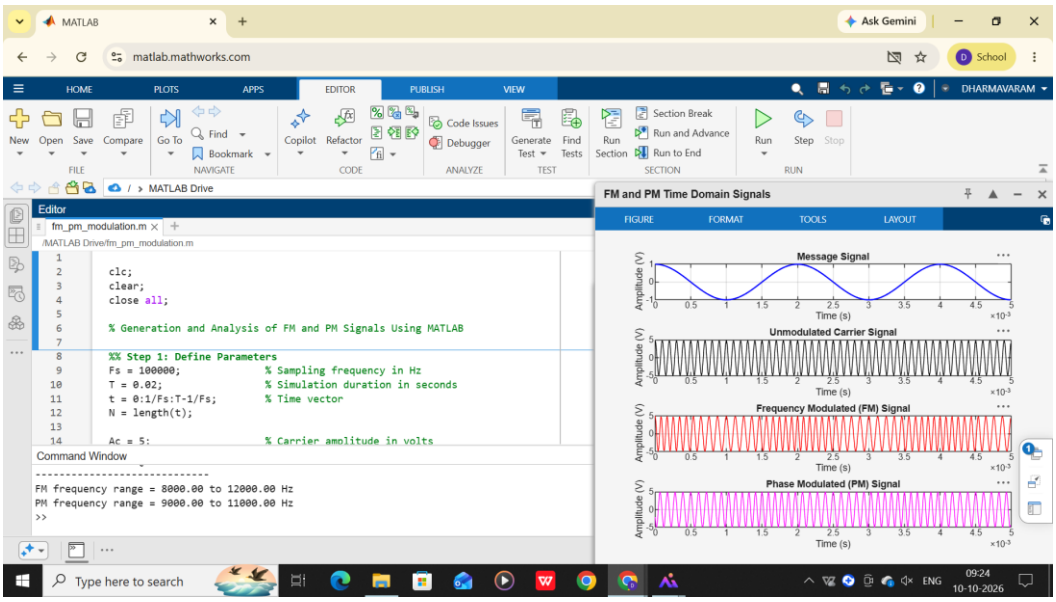
subplot(2,1,2); plot(t(1:n),fpm(1:n),'m'); grid on;
title('Instantaneous Frequency of PM'); xlabel('Time (s)'); ylabel('Frequency (Hz)');

f=(0:length(t)/2)*Fs/length(t);
FMsp=abs(fft(FM))/length(t);
PMsp=abs(fft(PM))/length(t);
```

```
FMsp=2*FMsp(1:length(f)); PMsp=2*PMsp(1:length(f));
figure;
subplot(2,1,1); plot(f/1000,FMsp,'r'); grid on;
title('FM Frequency Spectrum'); xlabel('Frequency (kHz)'); ylabel('Amplitude'); xlim([0 20]);
subplot(2,1,2); plot(f/1000,PMsp,'m'); grid on;
title('PM Frequency Spectrum'); xlabel('Frequency (kHz)'); ylabel('Amplitude'); xlim([0 20]);
```

Output:





Result: FM and PM signals were successfully generated and analyzed using MATLAB. Their waveforms, instantaneous frequency variations, and frequency spectra were observed and verified.